

Basic Statistical Methods

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
```

```
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times
library(broom) # Summarizes key information about statistical objects in tidy tibbles
library(broom.helpers) # Provides suite of functions to work with regression model 'broom::tidy()' t
```

Here are some R settings I use in this document:

```
rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE
```

Acknowledgements

This content is adapted from Vittinghoff et al. (2012), Chapter 3.

1 Introduction

This appendix reviews fundamental statistical methods that are prerequisites for the main content of this course. Most of this material should be familiar from Epi 202 and Epi 203.

Example dataset: HERS

Throughout this appendix we use the HERS dataset as a running example.

The “heart and estrogen/progestin study” (HERS) was a clinical trial of hormone therapy for prevention of recurrent heart attacks and death among 2,763 post-menopausal women with existing coronary heart disease (CHD) (Hulley et al. 1998).

The trial was conducted at 20 US clinical centers. Participants were randomized to receive either conjugated equine estrogens (0.625 mg/day) plus medroxyprogesterone acetate (2.5 mg/day) or a

matching placebo (Hulley et al. 1998). Women were followed for an average of 4.1 years (Hulley et al. 1998).

The primary outcome was nonfatal myocardial infarction or CHD death (Hulley et al. 1998).

```
library(haven)
hers <- haven::read_dta(
  paste0(
    "https://regression.ucsf.edu/sites/g/files",
    "/tkssra6706/f/wysiwyg/home/data/hersdata.dta"
  )
)
```

2 Descriptive Statistics

See Vittinghoff et al. (2012), §3.2.

2.1 Summary statistics for continuous variables

2.1.1 Sample mean

Definition 2.1 (Sample mean). The **sample mean** of n observations $x_1, \dots, x_n$ is:

$$\bar{x} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n x_i$$

2.1.2 Sample variance

Definition 2.2 (Sample variance). The **sample variance** is:

$$s^2 \stackrel{\text{def}}{=} \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

The divisor $n-1$ (rather than n) makes s^2 an unbiased^a estimator of the population variance σ^2 .

^a[estimation.qmd#def-unbiased](#)

2.1.3 Sample standard deviation

Definition 2.3 (Sample standard deviation). The **sample standard deviation** is $s \stackrel{\text{def}}{=} \sqrt{s^2}$. It is expressed in the same units as the original data, making it more interpretable than the variance.

2.1.4 Sample median

Definition 2.4 (Sample median). The **sample median** is the middle value when observations are sorted in ascending order. For n observations:

- If n is odd, the median is the $\frac{n+1}{2}$ th order statistic.
- If n is even, the median is the average of the $\frac{n}{2}$ th and $\frac{n}{2} + 1$ th order statistics.

The median is more robust to outliers than the mean.

2.1.5 Interquartile range

Definition 2.5 (Interquartile range). The **interquartile range (IQR)** is the difference between the 75th percentile (the third quartile, Q_3) and the 25th percentile (the first quartile, Q_1):

$$\text{IQR} \stackrel{\text{def}}{=} Q_3 - Q_1$$

Like the median, the IQR is robust to outliers.

2.2 Summary statistics for categorical variables

2.2.1 Sample proportion

Definition 2.6 (Sample proportion). For a binary outcome, the **sample proportion** of “successes” (coded as 1) is:

$$\hat{p} \stackrel{\text{def}}{=} \frac{k}{n}$$

where k is the number of successes and n is the total sample size.

2.3 Computing summary statistics in R

The `tbl_summary()` function from the `gtsummary` package produces formatted summary tables:

2.3.1 HERS baseline summary statistics

Exm

Example 2.1.

```

library(dplyr)
library(gtsummary)
library(haven)

hers |>
  mutate(
    HT = haven::as_factor(HT),
    exercise = haven::as_factor(exercise),
    smoking = haven::as_factor(smoking),
    diabetes = haven::as_factor(diabetes)
  ) |>
  select(age, BMI, glucose, SBP, DBP, HT, exercise, smoking, diabetes) |>
  tbl_summary(
    by = HT,
    statistic = list(
      gtsummary::all_continuous() ~ "{mean} ({sd})",
      gtsummary::all_categorical() ~ "{n} ({p}%)"
    ),
    digits = gtsummary::all_continuous() ~ 1,
    label = list(
      age ~ "Age (years)",
      BMI ~ "BMI (kg/m2)",
      glucose ~ "Fasting glucose (mg/dL)",
      SBP ~ "Systolic BP (mmHg)",
      DBP ~ "Diastolic BP (mmHg)",
      exercise ~ "Exercises regularly",
      smoking ~ "Current smoker",
      diabetes ~ "Diabetes"
    )
  ) |>
  add_overall() |>
  bold_labels()

```

Table 1

Characteristic	Overall N = 2,763 [†]	0 N = 1,383 [†]	1 N = 1,380 [†]
Age (years)	66.6 (6.7)	66.8 (6.7)	66.5 (6.6)
BMI (kg/m ²)	28.6 (5.5)	28.5 (5.5)	28.6 (5.5)
Unknown	5	4	1
Fasting glucose (mg/dL)	112.2 (36.8)	112.4 (36.8)	111.9 (36.9)
Systolic BP (mmHg)	135.1 (19.0)	135.1 (19.4)	135.0 (18.7)
Diastolic BP (mmHg)	73.2 (9.7)	73.1 (9.7)	73.2 (9.7)
Unknown	1	1	0
Exercises regularly			
0	1,695 (61%)	853 (62%)	842 (61%)
1	1,068 (39%)	530 (38%)	538 (39%)
Current smoker			
0	2,403 (87%)	1,201 (87%)	1,202 (87%)
1	360 (13%)	182 (13%)	178 (13%)
Diabetes			
0	2,032 (74%)	1,031 (75%)	1,001 (73%)
1	731 (26%)	352 (25%)	379 (27%)

[†]Mean (SD); n (%)

2.4 Exploratory data analysis

Graphical summaries reveal aspects of the data distribution that summary statistics may miss, such as skewness, multimodality, and outliers.

2.4.1 Histograms

A **histogram** displays the distribution of a continuous variable by dividing the range of values into intervals (bins) and plotting the number or proportion of observations in each bin.

2.4.2 Distribution of fasting glucose

Exm

Example 2.2.

```
library(ggplot2)

hers |>
  ggplot() +
  aes(x = glucose) +
  geom_histogram(binwidth = 5, fill = "steelblue", color = "white") +
  labs(
    x = "Fasting glucose (mg/dL)",
    y = "Count"
  )
```

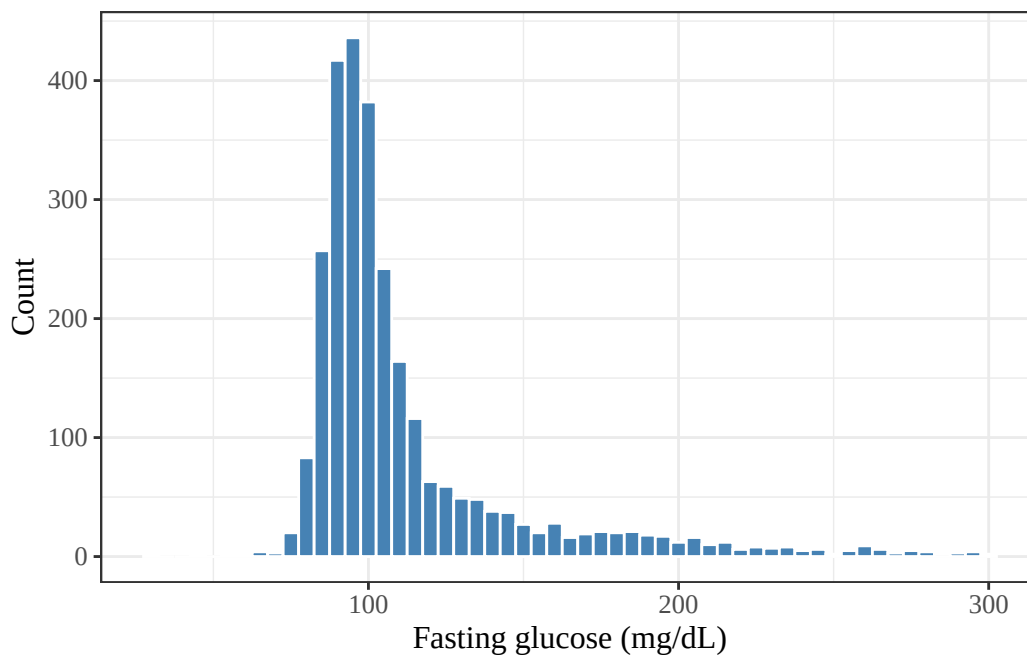


Figure 1: Distribution of fasting glucose (mg/dL) in HERS participants

2.4.3 Boxplots

A **boxplot** (box-and-whisker plot) summarizes the distribution of a continuous variable using five statistics: the minimum, the first quartile (Q_1), the median, the third quartile (Q_3), and the maximum (with outliers plotted separately).

2.4.4 Fasting glucose by hormone therapy group

Exm

Example 2.3.

```
hers |>
  mutate(HT = haven::as_factor(HT)) |>
  ggplot() +
  aes(x = HT, y = glucose) +
  geom_boxplot() +
  labs(
    x = "Hormone therapy",
    y = "Fasting glucose (mg/dL)"
  )
```

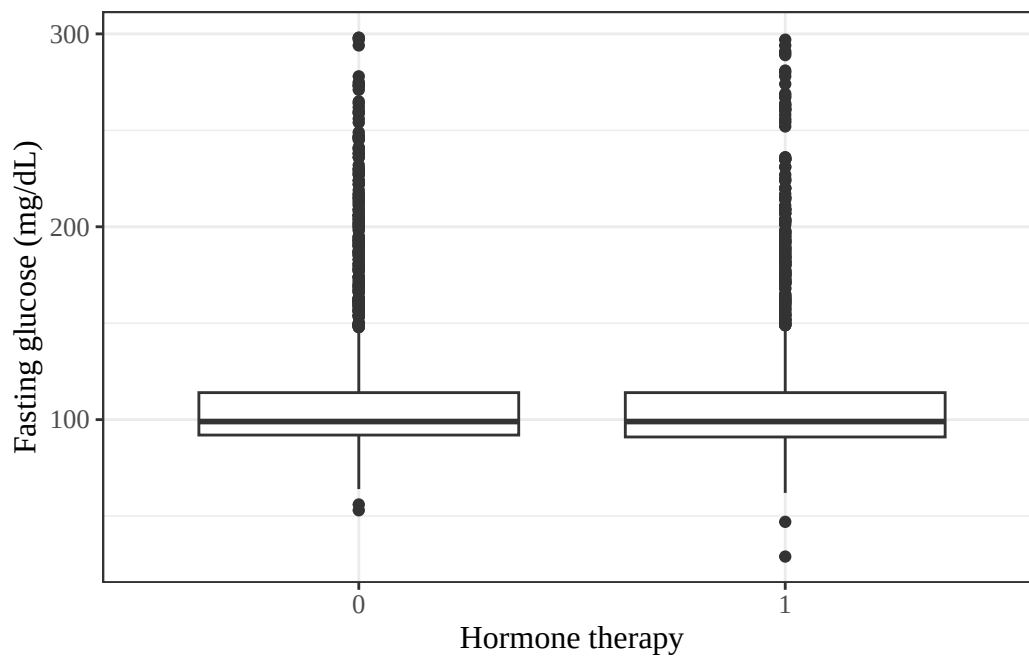


Figure 2: Fasting glucose by hormone therapy assignment in HERS

2.4.5 Scatterplots

A **scatterplot** displays the joint distribution of two continuous variables by plotting each observation as a point.

2.4.6 BMI versus fasting glucose

Exm

Example 2.4.

```
hers |>
  ggplot() +
  aes(x = BMI, y = glucose) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = TRUE) +
  labs(
    x = "BMI (kg/m2)",
    y = "Fasting glucose (mg/dL)"
  )
```

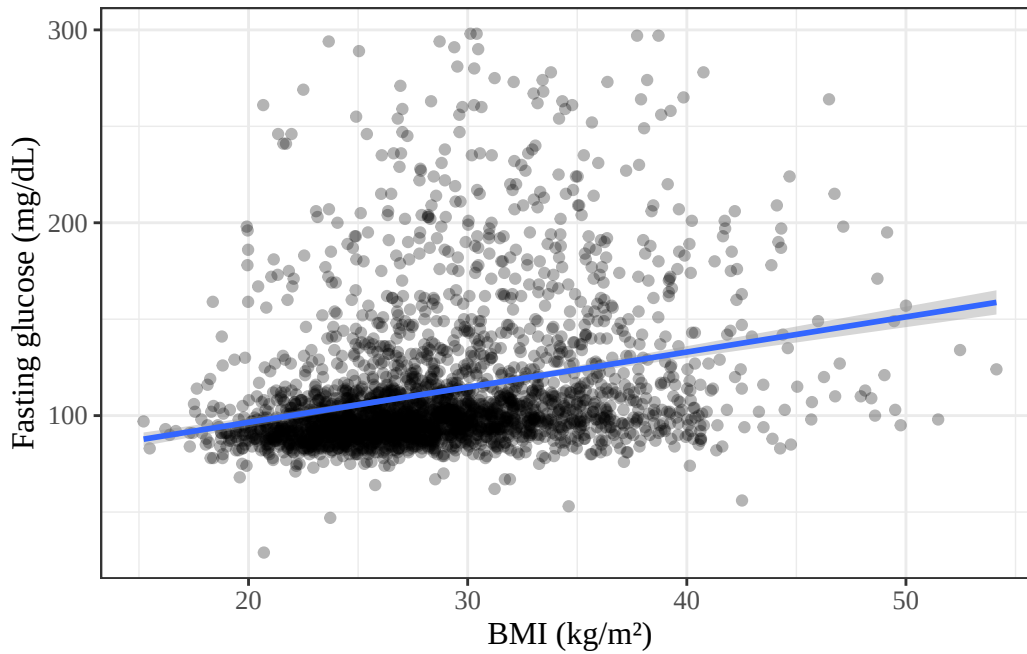


Figure 3: Fasting glucose versus BMI in HERS participants

3 Comparing Two Groups: Continuous Outcomes

See Vittinghoff et al. (2012), §3.3.

3.1 Hypotheses

3.1.1 Null hypothesis

Definition 3.1 (Null hypothesis). The **null hypothesis** H_0 is a specific claim about the population parameter(s) that we test against the data. In a two-group comparison of means, the null hypothesis is typically that the two group means are equal:

$$H_0 : \mu_1 = \mu_2$$

3.1.2 Alternative hypothesis

Definition 3.2 (Alternative hypothesis). The **alternative hypothesis** H_1 (or H_A) is the claim we are trying to find evidence for. For a two-sided test:

$$H_1 : \mu_1 \neq \mu_2$$

3.2 The two-sample t-test

3.2.1 Definition

Definition 3.3 (Two-sample t-test). The **two-sample t-test** (Welch's t-test) tests whether the means of two independent groups are equal.

For samples of sizes n_1 and n_2 from two groups with sample means $\bar{x}_1$, $\bar{x}_2$ and sample variances s_1^2 , s_2^2 , the test statistic is:

$$t \stackrel{\text{def}}{=} \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Under H_0 , this statistic follows (approximately) a t -distribution with degrees of freedom estimated by the Welch-Satterthwaite equation.

3.2.2 Welch's t-test vs. pooled t-test

i Note

Welch's t -test (the default in R's `t.test()`) does not assume equal variances across groups. The pooled t -test assumes equal variances ($\sigma_1^2 = \sigma_2^2$) and pools the two sample variances into a single estimate. Welch's t -test is generally preferred because the equal-variance assumption is rarely verifiable in practice (Vittinghoff et al. 2012, sec. 3.3).

3.2.3 Comparing fasting glucose between hormone therapy groups

Exm

Example 3.1. We test $H_0 : \mu_{\text{placebo}} = \mu_{\text{HT}}$ vs. $H_1 : \mu_{\text{placebo}} \neq \mu_{\text{HT}}$.

```
glucose_placebo <- hers |> filter(HT == 0) |> pull(glucose)
glucose_HT      <- hers |> filter(HT == 1) |> pull(glucose)

t.test(glucose_HT, glucose_placebo)
#>
#> Welch Two Sample t-test
#>
#> data: glucose_HT and glucose_placebo
#> t = -0.4246, df = 2761, p-value = 0.671
#> alternative hypothesis: true difference in means is not equal to 0
#> 95 percent confidence interval:
#>  -3.34503  2.15423
#> sample estimates:
#> mean of x mean of y
#>  111.854  112.449
```

3.3 One-sample t-test

3.3.1 Definition

Definition 3.4 (One-sample t -test). The **one-sample t -test** tests whether the mean of a single population equals a specified null value μ_0 :

$$H_0 : \mu = \mu_0 \quad H_1 : \mu \neq \mu_0$$

The test statistic is:

$$t \stackrel{\text{def}}{=} \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Under H_0 , $t \sim t_{n-1}$ (a t -distribution with $n - 1$ degrees of freedom).

3.4 Paired t-test

3.4.1 Definition

Definition 3.5 (Paired t-test). The **paired t-test** compares two related measurements (e.g., pre- and post-treatment values from the same subjects). Let $d_i \stackrel{\text{def}}{=} x_{i,1} - x_{i,2}$ be the within-subject difference; the test reduces to a one-sample t-test on the differences:

$$H_0 : \mu_d = 0 \quad H_1 : \mu_d \neq 0$$

3.4.2 Change in glucose over follow-up

Exm

Example 3.2.

```
# glucose1 is follow-up glucose; glucose is baseline
t.test(hers$glucose1, hers$glucose, paired = TRUE)
#>
#> Paired t-test
#>
#> data: hers$glucose1 and hers$glucose
#> t = 4.151, df = 2612, p-value = 3.42e-05
#> alternative hypothesis: true mean difference is not equal to 0
#> 95 percent confidence interval:
#>  1.38248 3.85824
#> sample estimates:
#> mean difference
#>      2.62036
```

3.5 Confidence intervals for the difference in means

A two-sided $(1 - \alpha) \cdot 100\%$ confidence interval for $\mu_1 - \mu_2$ is:

$$(\bar{x}_1 - \bar{x}_2) \pm t_{df}^* \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where t_{df}^* is the appropriate critical value from the t-distribution. The confidence interval is returned by `t.test()` in R alongside the hypothesis test result.

For more on confidence intervals, see Statistical Inference¹.

4 One-Way Analysis of Variance

Analysis of variance (ANOVA) generalizes the two-sample t-test to compare means across $k \geq 2$ groups.

4.1 Definition

Definition 4.1 (One-way ANOVA). In a **one-way ANOVA**, we test:

$$H_0 : \mu_1 = \mu_2 = \dots = \mu_k$$

vs. the alternative that at least one mean differs.

The F-statistic compares the **between-group variance** to the **within-group variance**:

¹[inference.qmd#sec-CI](#)

$$F \stackrel{\text{def}}{=} \frac{MS_{\text{between}}}{MS_{\text{within}}} = \frac{SS_{\text{between}}/(k-1)}{SS_{\text{within}}/(n-k)}$$

Under H_0 , $F \sim F_{k-1, n-k}$.

4.1.1 Fasting glucose by race/ethnicity

Exm

Example 4.1.

```
aov_result <- aov(glucose ~ factor(raceth), data = hers)
summary(aov_result)
#>               Df Sum Sq Mean Sq F value    Pr(>F)
#> factor(raceth)   2  45919   22959    17.1 4.1e-08 ***
#> Residuals      2760 3704543    1342
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4.1.2 ANOVA as a special case of linear regression

i Note

One-way ANOVA is equivalent to a linear regression model with a single categorical predictor. See Linear Models Overview^a for details.

^a[Linear-models-overview.qmd](#)

5 Comparing Two Groups: Categorical Outcomes

See Vittinghoff et al. (2012), §3.5.

5.1 Contingency tables

5.1.1 Definition

Definition 5.1 (Contingency table). A **contingency table** (cross-tabulation) displays the joint frequencies of two categorical variables. For two binary variables, this is a 2×2 table with cells a , b , c , d :

	Outcome = 1	Outcome = 0	Total
Exposure = 1	a	b	$a + b$
Exposure = 0	c	d	$c + d$
Total	$a + c$	$b + d$	n

5.1.2 Exercise by hormone therapy group

Exm

Example 5.1.

```

hers |>
  mutate(
    HT = haven::as_factor(HT),
    exercise = haven::as_factor(exercise)
  ) |>
  gtsummary::tbl_cross(
    row = exercise,
    col = HT,
    label = list(
      exercise ~ "Exercises regularly",
      HT ~ "Hormone therapy"
    ),
    percent = "row"
  )

```

Table 2: Exercise by hormone therapy group in HERS

	Hormone therapy		Total
	0	1	
Exercises regularly			
0	853 (50%)	842 (50%)	1,695 (100%)
1	530 (50%)	538 (50%)	1,068 (100%)
Total	1,383 (50%)	1,380 (50%)	2,763 (100%)

5.2 The chi-square test

5.2.1 Definition

Definition 5.2 (Chi-square test). The **Pearson chi-square test** tests whether two categorical variables are independent. For a 2×2 table, the test statistic is:

$$\chi^2 \stackrel{\text{def}}{=} \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is the observed cell count and $E_{ij} = \frac{(\text{row total}) \times (\text{column total})}{n}$ is the expected cell count under independence.

Under H_0 (independence), $\chi^2 \sim \chi_1^2$ for a 2×2 table.

5.2.2 Chi-square test: exercise vs. hormone therapy

Exm

Example 5.2.

```

chisq.test(hers$exercise, hers$HT)
#>
#> Pearson's Chi-squared test with Yates' continuity correction
#>
#> data:  hers$exercise and hers$HT
#> X-squared = 0.1016, df = 1, p-value = 0.75

```

5.3 Fisher's exact test

5.3.1 Definition

Definition 5.3 (Fisher's exact test). **Fisher's exact test** computes the exact probability of observing a 2×2 table at least as extreme as the observed table, given the marginal totals and under the null hypothesis of independence.

It is preferred over the chi-square test when cell counts are small (typically when any expected cell count is less than 5).

5.3.2 Fisher's exact test example

Exm

Example 5.3.

```
fisher.test(hers$exercise, hers$HT)
#>
#> Fisher's Exact Test for Count Data
#>
#> data:  hers$exercise and hers$HT
#> p-value = 0.725
#> alternative hypothesis: true odds ratio is not equal to 1
#> 95 percent confidence interval:
#>  0.879664 1.202192
#> sample estimates:
#> odds ratio
#>    1.02836
```

5.4 Measures of association for 2×2 tables

See Odds Ratios and Relative Risks² for definitions and formulas.

6 Correlation

See Vittinghoff et al. (2012), §3.6.

6.1 Pearson correlation coefficient

6.1.1 Definition

Definition 6.1 (Pearson correlation coefficient). The **Pearson correlation coefficient** measures the strength and direction of the linear association between two continuous variables X and Y :

$$r \stackrel{\text{def}}{=} \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}$$

r ranges from -1 (perfect negative linear relationship) to $+1$ (perfect positive linear relationship); $r = 0$ indicates no linear association.

²[binary-outcome-associations.qmd#sec-OR-RR](#)

6.1.2 Correlation between BMI and glucose

Exm

Example 6.1.

```
cor.test(hers$BMI, hers$glucose, method = "pearson")
#>
#> Pearson's product-moment correlation
#>
#> data:  hers$BMI and hers$glucose
#> t = 14.88, df = 2756, p-value <2e-16
#> alternative hypothesis: true correlation is not equal to 0
#> 95 percent confidence interval:
#>  0.237760 0.306865
#> sample estimates:
#>      cor
#> 0.272664
```

6.2 Spearman rank correlation

6.2.1 Definition

Definition 6.2 (Spearman rank correlation). The **Spearman rank correlation** r_S is the Pearson correlation computed on the *ranks* of the observations. It measures the strength and direction of any *monotone* association (not just linear) and is more robust to outliers.

6.2.2 Spearman correlation between BMI and glucose

Exm

Example 6.2.

```
cor.test(hers$BMI, hers$glucose, method = "spearman")
#>
#> Spearman's rank correlation rho
#>
#> data:  hers$BMI and hers$glucose
#> S = 2.33e+09, p-value <2e-16
#> alternative hypothesis: true rho is not equal to 0
#> sample estimates:
#>      rho
#> 0.333751
```

7 Simple Linear Regression

See Vittinghoff et al. (2012), §3.6 and Linear Models Overview³.

³[Linear-models-overview.qmd](#)

7.1 Model specification

7.1.1 Definition

Definition 7.1 (Simple linear regression). A **simple linear regression** model relates a continuous outcome Y to a single predictor X :

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2)$$

- β_0 is the **intercept**: the expected value of Y when $X = 0$.
- β_1 is the **slope**: the expected change in Y per one-unit increase in X .
- ε_i are independent Gaussian errors with mean 0 and variance σ^2 .

7.2 Ordinary least squares estimation

The parameters β_0 and β_1 are estimated by minimizing the **residual sum of squares (RSS)**:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

The closed-form **ordinary least squares (OLS)** estimators are:

$$\hat{\beta}_1 = r \cdot \frac{s_y}{s_x}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where r is the Pearson correlation and s_x , s_y are the sample standard deviations.

7.3 Fitting a simple linear regression in R

7.3.1 Glucose on BMI

Exm

Example 7.1.

Table 3

```
slr_fit <- lm(glucose ~ BMI, data = hers)
summary(slr_fit)
#>
#> Call:
#> lm(formula = glucose ~ BMI, data = hers)
#>
#> Residuals:
#>    Min       1Q   Median       3Q      Max
#> -81.55 -18.98 -10.35   3.76 190.81
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept)   60.074      3.565    16.9   <2e-16 ***
#> BMI           1.822      0.122    14.9   <2e-16 ***
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 35.5 on 2756 degrees of freedom
#> (5 observations deleted due to missingness)
#> Multiple R-squared:  0.0743, Adjusted R-squared:  0.074
#> F-statistic: 221 on 1 and 2756 DF, p-value: <2e-16
```

The estimated slope is $\hat{\beta}_1 = 1.82$ mg/dL per kg/m², meaning fasting glucose increases by approximately 1.82 mg/dL for each 1 kg/m² increase in BMI.

7.4 The coefficient of determination (R^2)

7.4.1 Definition

Definition 7.2 (Coefficient of determination (R^2)). The **coefficient of determination** R^2 measures the proportion of the total variance in Y that is explained by the linear regression on X :

$$R^2 \stackrel{\text{def}}{=} 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

R^2 ranges from 0 (no linear relationship) to 1 (perfect linear fit). For simple linear regression, $R^2 = r^2$.

7.5 Further reading

For a more thorough treatment of linear regression, see Linear Models Overview⁴ and Vittinghoff et al. (2012), Chapters 4 and 9.

8 Bootstrap Confidence Intervals

Adapted from (Vittinghoff et al. 2012, sec. 3.6, pp. 62–63).

8.1 When to use the bootstrap

Bootstrapping is a widely applicable method for obtaining standard errors and CIs in three situations:

- Approximate methods for computing valid CIs have been developed but are not conveniently implemented in standard software.
- Developing closed-form approximate methods has turned out to be intractable.
- The dataset violates the assumptions underlying established methods badly enough that the resulting CIs would be unreliable.

8.2 The bootstrap procedure

Definition 8.1 (Bootstrap sample). A **bootstrap sample** is a sample of size n drawn **with replacement** from the observed data of size n . Because sampling is with replacement, each bootstrap sample contains some observations more than once and omits others entirely.

Exm

Example 8.1 (Bootstrap sample). Consider a dataset of five observations: {3, 7, 5, 2, 9} ($n = 5$). Two possible bootstrap samples drawn with replacement are:

Bootstrap sample	Values drawn	Notes
1	3, 3, 7, 5, 9	3 appears twice; 2 not drawn
2	2, 7, 7, 9, 5	7 appears twice; 3 not drawn

⁴[Linear-models-overview.qmd](#)

Each sample has size $n = 5$ but may repeat or omit individual observations from the original dataset.

Definition 8.2 (Bootstrap distribution). The **bootstrap distribution** of a statistic is the empirical distribution of that statistic computed across a large number B of bootstrap samples. It serves as an estimate of the sampling distribution of the statistic. The **bootstrap standard error** is the standard deviation of the bootstrap distribution.

Exm

Example 8.2 (Bootstrap distribution).

```
set.seed(42)
toy <- c(3, 7, 5, 2, 9, 4, 8, 1, 6, 10)
boot_means_toy <- replicate(
  1000,
  mean(sample(toy, replace = TRUE))
)
boot_se_toy <- sd(boot_means_toy)
boot_ci_norm_lo <- mean(toy) - 1.96 * boot_se_toy
boot_ci_norm_hi <- mean(toy) + 1.96 * boot_se_toy
boot_ci_perc <- quantile(boot_means_toy, c(0.025, 0.975))
```

Using the 10-observation dataset $\{3, 7, 5, 2, 9, 4, 8, 1, 6, 10\}$ (observed mean 5.5), we draw $B = 1,000$ bootstrap resamples and compute the sample mean of each. The bootstrap distribution is centered near the observed mean with bootstrap standard error 0.92.

```
library(ggplot2)
ggplot(data.frame(mean = boot_means_toy)) +
  aes(x = mean) +
  geom_histogram(bins = 30, color = "white", fill = "steelblue") +
  geom_vline(xintercept = mean(toy), color = "red", linewidth = 1) +
  labs(x = "Bootstrap mean", y = "Count")
```

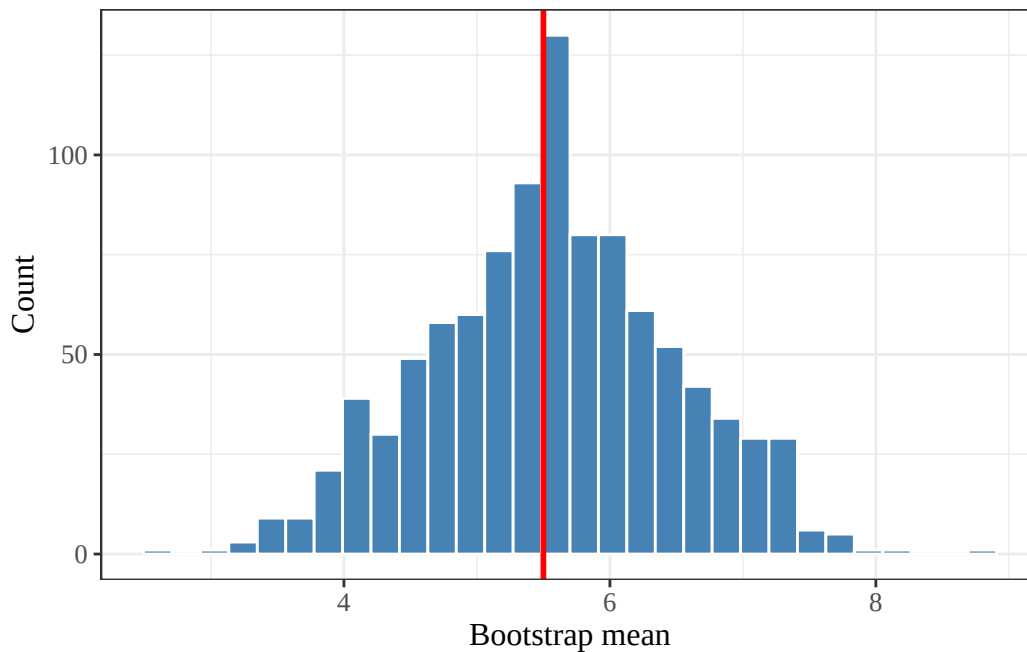


Figure 4: Bootstrap distribution of the sample mean ($B = 1,000$ replicates, $n = 10$ toy dataset). The red line marks the observed mean (5.5). The distribution is approximately normal and centered near $\bar{x}$.

The key idea is to treat the observed sample as a stand-in for the source population, and to mimic repeated sampling from that population by resampling with replacement from the observed data.

8.3 Bootstrap CI methods

Three methods for constructing a CI from the bootstrap distribution are in common use (Vittinghoff et al. 2012, sec. 3.6, p. 63):

8.3.1 Normal approximation

If the bootstrap distribution of the statistic of interest is approximately normal, replace the model-based standard error in a conventional CI with the bootstrap standard deviation:

$$\hat{\theta} \pm z^* \cdot \widehat{SE}_{\text{boot}}$$

This method requires fewer bootstrap replicates than percentile-based methods, but is less reliable when the sampling distribution departs from normality, particularly in the tails.

Exm

Example 8.3 (Normal approximation). Continuing from Example 8.2: the observed mean is 5.5 and the bootstrap standard error is 0.92. The 95% normal-approximation CI is:

$$\bar{x} \pm 1.96 \cdot \widehat{SE}_{\text{boot}}$$

which here evaluates to (3.71, 7.29).

8.3.2 Percentile method

The CI is constructed from the empirical quantiles of the bootstrap distribution. A 95% CI spans the 2.5th to 97.5th percentiles of the B bootstrap estimates.

Because the extreme percentiles of a finite sample are noisy estimates of the corresponding population quantiles, a larger number of replicates is required (typically $B \geq 1,000$).

Exm

Example 8.4 (Percentile method). Continuing from Example 8.2 with $B = 1,000$ bootstrap means: the 2.5th percentile is 3.8 and the 97.5th percentile is 7.3, giving a 95% CI of (3.8, 7.3).

8.3.3 Bias-corrected and accelerated (BCa) method

The BCa interval adjusts the percentile-based CI to account for both bias (a shift between the observed statistic and the median of the bootstrap distribution) and skewness in the bootstrap distribution (James et al. 2021, chap. 5, p. 209). BCa intervals are preferred when the bootstrap distribution is noticeably skewed.

Exm

Example 8.5 (Bias-corrected and accelerated method). The BCa interval for the toy dataset is computed automatically by `boot.ci()` in R, which handles the bias-correction and acceleration adjustments internally. The worked example in Section 8.4.1 demonstrates all three CI methods side by side, making it straightforward to compare the BCa interval to the normal and percentile intervals.

8.4 Bootstrap CI in R

The `boot` package (included with base R) provides `boot()` for resampling and `boot.ci()` for all three CI types.

8.4.1 Bootstrap CI for the slope of SBP on age

Exm

Example 8.6. We adapt the example from (Vittinghoff et al. 2012, sec. 3.6, p. 62): a simple linear regression of systolic blood pressure (SBP) on `age` in the HERS dataset.

```

library(boot)

# statistic function: returns the slope of SBP ~ age
boot_slope <- function(data, indices) {
  fit <- lm(SBP ~ age, data = data[indices, ])
  coef(fit)[["age"]]
}

set.seed(42)
boot_result <- boot(
  data = hers,
  statistic = boot_slope,
  R = 1000
)

boot_result
#>
#> ORDINARY NONPARAMETRIC BOOTSTRAP
#>
#>
#> Call:
#> boot(data = hers, statistic = boot_slope, R = 1000)
#>
#>
#> Bootstrap Statistics :
#>      original      bias    std. error
#> t1* 0.471728 -0.000635537  0.0536514

```

The bootstrap standard error closely matches the model-based standard error from `lm()`. All three bootstrap intervals reported by `boot.ci()` are consistent with the parametric CI.

```

boot.ci(
  boot_result,
  type = c("norm", "perc", "bca")
)
#> BOOTSTRAP CONFIDENCE INTERVAL CALCULATIONS
#> Based on 1000 bootstrap replicates
#>
#> CALL :
#> boot.ci(boot.out = boot_result, type = c("norm", "perc", "bca"))
#>
#> Intervals :
#> Level      Normal          Percentile          BCa
#> 95%   ( 0.3672, 0.5775 )   ( 0.3607, 0.5811 )   ( 0.3624, 0.5824 )
#> Calculations and Intervals on Original Scale

```

The normal, percentile, and BCa intervals are all similar here, consistent with the bootstrap distribution being approximately symmetric. In cases with more skewness, the BCa interval would differ more from the others, and should be preferred.

Hulley, Stephen, Deborah Grady, Trudy Bush, et al. 1998. "Randomized Trial of Estrogen Plus Progestin for Secondary Prevention of Coronary Heart Disease in Postmenopausal Women." *JAMA : The Journal of the American Medical Association* (Chicago, IL) 280 (7): 605–13.

James, Gareth, Daniela Witten, Trevor Hastie, and Robert Tibshirani. 2021. *An Introduction to Statistical Learning: With Applications in R*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-0716-1418-1>.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression*

Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models. 2nd ed.
Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.